CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student concludes that 'all metals are heavy' after testing iron and lead. This is: 1

(a) an over-generalisation from few examples **(b)** a correct conclusion
(c) a hypothesis **(d)** a law

2. Which hypothesis is testable? 1

(a) Salt is good.
(b) Salt dissolves faster in hot water than in cold water.
(c) Hot water is nicer.
(d) Salt tastes better than sugar.

3. A student reads a thermometer with the eye above the scale. The error is: 1

(a) parallax error **(b)** time error
(c) temperature error **(d)** zero error

4. The best way to display marks scored by 5 students in a test is: 1

(a) a map **(b)** a bar graph
(c) a pie chart of time **(d)** a line graph of marks against names

5. A table has columns for temperature (degree C) and time (s). The best title for the second column is: 1

(a) Seconds taken by me **(b)** Time
(c) Measurement **(d)** Time taken (s)

6. Which of these is an inference? 1

(a) The soil is dark. **(b)** The soil weighs 200 g.
(c) The soil is at 25 degrees C. **(d)** The soil is wet, so it rained.

7. Which of the following is NOT needed to repeat an experiment? 1

(a) method **(b)** recorded observations
(c) personal opinion of the investigator **(d)** apparatus used

8. An outlier in a data set is: 1

(a) the average **(b)** a conclusion
(c) the largest unit **(d)** a value far from the others

9. The control in an experiment is used to: 1

(a) avoid the result
(b) reduce time
(c) compare the changed set with an unchanged set
(d) waste material

10. Taking readings in a hurry and writing from memory causes: 1

(a) better results **(b)** a longer experiment
(c) greater precision **(d)** recording errors

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a flame source not kept near inflammable liquids? 2

12. Explain the term 'zero error' in a measuring instrument. 2

13. Give two reasons why results should be written in a table before drawing a graph. 2

14. State two uses of a line graph in an investigation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why should an investigator not rely on a single experiment to make a general claim? Explain with an example. 3

16. Name three kinds of graphs and mention a use for each. 3

17. Explain what is meant by 'the investigation led to a new question'. 3

OR

Explain how a graph can reveal an error in the data. 3

18. Compare a hypothesis and a prediction. 3

OR

A student says 'plants need light because they grew taller near the window'. Is this a valid conclusion? Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student kept three identical plants: one in sunlight, one in shade and one in a dark cupboard, each given 50 mL water daily. Identify the independent, dependent variables and constants and the expected result. 5

OR

How can the use of the scientific method in daily life help a person make better decisions? Give two examples. 5

20. Describe how errors may arise in using a thermometer and how to avoid them. 5

OR

Explain with an example why a hypothesis that is rejected can still be useful. 5

END OF QUESTION PAPER — Total: 40 Marks